

Running: reward speed without rewarding a shuffle

01 Input

Observable forward command 0.75 m/s; nominal robot



02 Acquire

Fresh PPO base; require both-feet aerial phases



03 Refine

Yaw-rate tracking reduces circular reward exploitation



04 Evaluate

12 seconds; displacement ≥ 2 m; aerial fraction $\geq 3\%$



05 Output

ONNX + speed/contact traces + full physical rollout

All measured outcomes are nominal simulation evidence; proposed extensions are untested.